

Cal Hargis

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<https://calhargis.github.io>

EXPERIENCE

Fullstack Software Developer

Apr 2025-Present

Kingsley Management Corporation

Provo, UT

- Designed and implemented full-stack features using Laravel (PHP), Vue.js, MySQL, and Flutter, owning frontend UX, backend APIs, and database architecture to deliver production functionality for internal management tools and resident-facing applications.
- Built and maintained RESTful backend services using Laravel Controllers, Models, and Service layers, ensuring clean separation of concerns and long-term maintainability.
- Developed features across internal property management systems and resident-facing web portals.
- Led development of an online security deposit and refund system, replacing a legacy paper-check workflow with a fully digital solution and significantly improving processing speed and reliability while processing over \$100,000 monthly.
- Implemented and managed relational database schemas and queries in MySQL, ensuring data integrity across financial, resident, and property records.

Application Support Analyst

Jul 2024-Apr 2025

JobNimbus

Lehi, UT

- Diagnosed and resolved complex technical issues across web and mobile platforms, frequently involving database records, application logic, and API behavior.
- Collaborated with engineering and product teams to reproduce bugs, validate fixes, and improve system reliability based on real customer usage.

Software Engineer Intern

May 2023-Dec 2023

Family Search

Lehi, UT

- Constructed robust production code for FamilySearch, leveraging Java Apache Spark and Spring Boot, to efficiently extract hint files from a large-scale AWS database, improving data extraction processes.
- Developed Java scripts to efficiently extract over 2M+ hint files from AWS servers utilizing AWS CloudFormation, substantially enhancing data retrieval processes.
- Utilized AWS monitoring and logging tools to track and optimize the performance of data extraction processes.
- Produced bash scripts that optimized AWS CloudFormation processes and reduced run times by 83%.

Research Assistant

Apr 2020-Dec 2022

Brigham Young University

Provo, UT

- Engineered robust data pipelines leveraging NumPy, Pandas, and Scikit-learn; conducted statistical analyses and developed machine learning models, improving predictive accuracy by 25% and reducing data processing time by 40%.
- Assisted in 2 research papers and presentations by providing visualizations generated from random forest model analyses.

EDUCATION

BS in Computer Science

Apr 2024

Brigham Young University

Provo, UT

- Relevant Coursework: Software Engineering, Android Development, UX Design, Software Design & Testing, Algorithms, Machine Learning.

PROJECTS/RESEARCH/PRESENTATION

Machine Learning Analysis of Dynamic-Dependent Bond Formation in Trajectories with Consecutive Transition States

- Published in Wiley Online Library, a reputable scientific journal with over 8 million articles from 1,600 journals.
- Presented at the 2022 BYU Student Research Conference on using machine learning to uncover the origin of dynamic selectivity in sigmatropic rearrangement reactions through quasiclassical direct dynamics simulations.

Machine Learning Classification of Disrotatory IRC and Conrotatory Non-IRC Trajectory Motion for Cyclopropyl Radical Ring Opening

- Co-authored and published in Royal Society of Chemistry, the oldest and one of the most highly respected journals in Chemistry, demonstrating expertise in machine learning and data analysis.

SKILLS

- Languages & Frameworks: PHP (Laravel), JavaScript (Vue.js), Flutter (Dart), SQL (MySQL), Java, C++, Python
- Web & Backend: REST APIs, MVC Architecture, Service Layer Design, Authentication, Data Modeling
- Databases & Cloud: MySQL, AWS, SQL Schema Design
- Tools & Practices: Git, Agile/Scrum, Linux, AI Interfaces (Codex, Claude Code, OpenClaw)